

WardSync Backend — Integration Guide

What is WardSync?

WardSync is a real-time field hospital management system. It handles patient triage, vital signs monitoring, treatment records, medication inventory, and room capacity tracking. The backend is built with **Express.js + TypeScript + Firebase (Firestore, Auth, Storage, FCM)**.

This document is written for the Flutter frontend team. If you are using an AI assistant to help you integrate, share this entire file as context.

Base URLs

Environment	URL
Local (emulator)	http://localhost:3000
Production	TBD

Authentication

Every request (except login) requires a Firebase ID Token in the Authorization header:

```
Authorization: Bearer <idToken>
```

How to get a token (Login)

```
POST
http://localhost:9099/identitytoolkit.googleapis.com/v1/accounts:signInWithPassword?
key=fake-key
```

```
Body (JSON):
{
  "email": "user@example.com",
  "password": "password123",
  "returnSecureToken": true
}
```

```
Response:  
{  
  "idToken": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXZWQ..."} ← use this as Bearer token
```

In production, use the real Firebase Auth SDK in Flutter (`signInWithEmailAndPassword`). The `idToken` is obtained via `user.getIdToken()` .

First-time login flow

1. User logs in → gets `idToken`
2. Call `GET /api/users/profile`

3. If response has `isProfileComplete: false` → redirect to profile setup screen
4. User fills in name, role, assigned room → `PATCH /api/users/profile`
5. After profile is complete → proceed to main app

Roles & Permissions

There are 3 roles: `nurse` , `doctor` , `admin` .

Endpoint / Action	nurse	doctor	admin
View own profile	✓	✓	✓
Complete profile	✓	✓	✓
Update FCM token	✓	✓	✓
List all users	✗	✗	✓
Create user account	✗	✗	✓
Change user role	✗	✗	✓
Create patient	✓	✗	✗
List patients	✓ (all rooms)	✓ (own room only)	✓ (all rooms)
View patient detail	✓	✓	✓
Change patient status	✗	✓	✗
Add / edit vitals	✓	✓	✗
Add / edit treatments	✗	✓	✗
View vitals & treatments	✓	✓	✓
View medication list	✓	✓	✓
Manage medication inventory	✗	✗	✓
View room capacity	✓	✓	✓
Set room capacity	✗	✗	✓
View statistics charts	✗	✗	✓

Doctor room filter: Doctors automatically see only patients in their assigned room. To see all rooms, add `?room=all` .

Nurse: Does not have an `assignedRoom` . Can see all patients across all rooms.

Error Response Format

All errors follow this format:

```
{
  "statusCode": 404,
  "error": "Not Found",
  "message": "Patient not found"
}
```

Status Code	Meaning
400	Bad Request — invalid or missing fields
401	Unauthorized — missing or expired token
403	Forbidden — role does not have permission
404	Not Found
409	Conflict — e.g. wristband number already in use

User Management (F1)

Get my profile

GET /api/users/profile

Response:

```
{
  "uid": "abc123",
  "email": "nurse@hospital.com",
  "displayName": "Nurse Anna",
  "role": "nurse",
  "assignedRoom": null,
  "isProfileComplete": true,
  "fcmToken": "...
}
```

Complete / update profile

PATCH /api/users/profile

Body:

```
{
  "displayName": "Dr. Smith",
  "role": "doctor",
  "assignedRoom": "red"
}
```

`assignedRoom` is required for `doctor` and `admin`. Valid values: `red`, `yellow`, `green`, `black`.
`nurse` does not need `assignedRoom`.

Update FCM token (call this on every app launch)

```
PATCH /api/users/fcm-token
```

Body:

```
{ "fcmToken": "<token from Firebase Messaging>" }
```

Admin: List all users

```
GET /api/users
```

Admin: Create a new user account

```
POST /api/users
```

Body:

```
{
  "email": "doctor@hospital.com",
  "password": "initial123",
  "displayName": "Dr. Smith",
  "role": "doctor",
  "assignedRoom": "red"
}
```

Admin: Change user role

```
PATCH /api/users/:uid/role
```

Body:

```
{
  "role": "doctor",
  "assignedRoom": "yellow"
}
```

Patient Management (F2, F3, F5)

Create patient (nurse only) — multipart/form-data

```
POST /api/patients
```

Content-Type: multipart/form-data

Fields:

wristbandNumber	string	e.g. "003"
triageColor	string	red yellow green black
sex	string	male female
ageRange	string	infant child adult elder

File (optional):

photo	image file (jpg/png)
-------	----------------------

Response: Patient object (see below)

Patient object structure

```
{
  "id": "abc123",
  "wristbandNumber": "003",
  "triageColor": "red",
  "sex": "male",
  "ageRange": "adult",
  "status": "waiting",
  "assignedRoom": "red",
  "photoUrl": "https://storage.../photo.jpg",
  "statusHistory": [
    { "status": "waiting", "changedBy": "uid", "changedAt": "timestamp" }
  ],
  "createdBy": "uid",
  "createdAt": "timestamp",
  "updatedAt": "timestamp"
}
```

List patients (with filters)

GET /api/patients

Query parameters (all optional, can be combined):

room=red|yellow|green|black|all

status=waiting|inTreatment|discharged|deceased

triageColor=red|yellow|green|black

wristband=00 ← prefix search, e.g. finds "001", "003", "009"

today=true ← only patients admitted today

fromTime=2026-06-01T08:00:00Z

toTime=2026-06-01T17:00:00Z

Response:

```
{
  "patients": [ ...Patient objects ],
  "total": 10
}
```

Get single patient

GET /api/patients/:id

Change patient status (doctor only)

PATCH /api/patients/:id/status

Body:

```
{ "status": "inTreatment" }
```

Valid status flow: waiting → inTreatment → discharged | deceased

Admin: Get patient counts summary

GET /api/patients/stats

Response:

```
{
  "total": 10,
  "byColor": { "red": 5, "yellow": 3, "green": 2, "black": 0 },
  "byRoom": { "red": 5, "yellow": 3, "green": 2, "black": 0 },
  "byStatus": { "waiting": 4, "inTreatment": 3, "discharged": 2, "deceased": 1 }
}
```

Vital Signs (F5, F7)

Add vitals

POST /api/patients/:id/vitals

Body:

```
{
  "bloodPressure": "120/80",
  "heartRate": 75,
  "temperature": 37.0,
  "oxygenSaturation": 98,
  "respiratoryRate": 16
}
```

Response includes an "alerts" field:

```
{
  "id": "vitalId",
  "heartRate": 30,
  ...
  "alerts": {
    "heartRate": "HR critical: 30 bpm"
  }
}
```

If `alerts` is not empty `{}`, highlight those fields in **red** in the UI.

A FCM push notification is also automatically sent to the doctor in the same room.

Critical thresholds (auto-checked on save)

Vital	Critical Range
Blood Pressure	systolic < 90 or > 180 / diastolic < 60 or > 120

Heart Rate	< 40 or > 150 bpm
Temperature	< 35 or > 39.5 °C
SpO2	< 90%
Respiratory Rate	< 8 or > 30 /min

List vitals

```
GET /api/patients/:id/vitals
```

Update vitals (partial update — only send fields you want to change)

```
PATCH /api/patients/:id/vitals/:vitalId
```

Body (all fields optional):

```
{
  "heartRate": 85,
  "temperature": 37.5
}
```

Delete vitals

```
DELETE /api/patients/:id/vitals/:vitalId
```

Treatments (F5)

Add treatment (doctor only)

```
POST /api/patients/:id/treatments
```

Body:

```
{
  "diagnosis": "Suspected sepsis",
  "treatment": "Fluid resuscitation",
  "notes": "Monitor BP every 15 min",
  "medications": [
    { "medicationId": "<id from medication list>", "dosage": "500ml IV bolus" }
  ]
}
```

`medications` is optional. When provided, stock is automatically decremented.

If a medication is `outOfStock`, the request will be rejected with a 400 error.

List treatments

```
GET /api/patients/:id/treatments
```

Response: array of Treatment objects

```
{
  "id": "...",
  "diagnosis": "...",
  "treatment": "...",
  "notes": "...",
  "medications": [
    { "medicationId": "...", "name": "Normal Saline", "dosage": "500ml IV bolus" }
  ],
  "recordedBy": "uid",
  "recordedAt": "timestamp"
}
```

Update treatment

PATCH /api/patients/:id/treatments/:treatmentId

Body (all optional):

```
{ "diagnosis": "Updated diagnosis", "notes": "New note" }
```

Delete treatment

DELETE /api/patients/:id/treatments/:treatmentId

Push Notifications (F4)

Backend sends FCM automatically. Flutter must:

1. Request notification permission from user
2. Get FCM token: `FirebaseMessaging.instance.getToken()`
3. Send token to backend: PATCH /api/users/fcm-token
4. Refresh token when it rotates: `FirebaseMessaging.instance.onTokenRefresh`

Notification types and payload

type in data	Trigger	Navigate to
new_patient	New patient admitted in doctor's room	Patient detail screen (use patientId)
vital_alert	Critical vital sign recorded	Patient vitals screen (use patientId)
low_stock	Medication stock $\leq$ threshold	Medication inventory screen (use medicationId)
out_of_stock	Medication quantity = 0	Medication inventory screen (use medicationId)

Medication Inventory (F8)

List medications (all roles)

```
GET /api/medications
GET /api/medications?search=saline    ← search by name

Response: array of:
{
  "id": "...",
  "name": "Normal Saline",
  "dosage": "500ml IV bolus",
  "quantity": 45,
  "lowStockThreshold": 10,
  "status": "inStock",    ← inStock | lowStock | outOfStock
  "createdAt": "...",
  "updatedAt": "..."
}
```

Get single medication

```
GET /api/medications/:id
```

Admin: Create medication

```
POST /api/medications

Body:
{
  "name": "Normal Saline",
  "dosage": "500ml IV bolus",
  "quantity": 100,
  "lowStockThreshold": 10
}
```

`lowStockThreshold` is optional, defaults to 10.

Admin: Update medication details

```
PATCH /api/medications/:id

Body (all optional):
{
  "name": "Updated Name",
  "dosage": "250ml IV",
  "lowStockThreshold": 15
}
```

Admin: Update stock count

```
PATCH /api/medications/:id/stock
```

Body:

```
{ "delta": 50 }    ← positive = add stock, negative = remove stock
```

Response: updated Medication object

Admin: Delete medication

```
DELETE /api/medications/:id
```

Room Capacity (F9)

Get all rooms capacity

```
GET /api/rooms/capacity
```

Response: array of:

```
{
  "room": "red",
  "current": 8,
  "maxCapacity": 10,
  "percentage": 80,
  "warning": "nearFull"    ← normal | nearFull | full
}
```

Warning	Condition	Display
normal	< 80%	No indicator
nearFull	≥ 80%	Yellow warning
full	≥ 95%	Red warning

Get single room

```
GET /api/rooms/capacity/:room
```

Admin: Set room capacity

```
PUT /api/rooms/capacity/:room
```

Body:

```
{ "maxCapacity": 15 }
```

Statistics Charts (F11)

```
GET /api/stats/charts          ← default: last 7 days
GET /api/stats/charts?days=30
GET /api/stats/charts?fromDate=2026-06-01&toDate=2026-06-07
```

Response:

```
{
  "patientsPerDay": [
    { "date": "2026-05-26", "count": 3 },
    { "date": "2026-05-27", "count": 5 },
    ...
  ],
  "byColor": {
    "red": 9, "yellow": 3, "green": 5, "black": 1
  },
  "mortalityRate": [
    { "date": "2026-05-26", "deceased": 1, "total": 3, "rate": 0.333 }
  ],
  "avgResponseTimeMinutes": 54.5
}
```

All dates are in **UTC**. Convert to local time before displaying.

Frontend Integration Checklist

Complete these in order before testing any feature:

- ☐ Add `google-services.json` (Android) / `GoogleService-Info.plist` (iOS) to Flutter project
- ☐ Implement login screen → call Firebase Auth → store `idToken`
- ☐ On every app launch: `GET /api/users/profile`
 - If `isProfileComplete: false` → show profile setup screen
 - If complete → route by role (`nurse` , `doctor` , `admin`)
- ☐ After profile check: get FCM token → `PATCH /api/users/fcm-token`
- ☐ Set up FCM `onMessage` and `onMessageOpenedApp` handlers (see Notification Types above)
- ☐ Before showing treatment form: fetch `GET /api/medications` for the medication picker
- ☐ When displaying vitals: check `alerts` field → show critical fields in red
- ☐ When displaying room status: map `warning` field to UI color (`normal` =green, `nearFull` =yellow, `full` =red)
- ☐ Admin stats screen: call `GET /api/stats/charts?days=7` and map to chart widgets